

WHO KNOWS WHO LIVES HERE?

WITHIN-HOUSEHOLD DISAGREEMENTS AS A SOURCE OF SURVEY COVERAGE ERROR

ELIZABETH MARTIN

Abstract Despite evidence of coverage errors and net undercounts in surveys due to listing errors and omissions from household rosters, little systematic attention has been devoted to the measurement properties of household rosters. This article draws on data from the Living Situation Survey, a pilot study that experimented with expanded roster cues and probes to identify individuals with marginal or tenuous attachments to households. The data are analyzed to test hypotheses that disagreements among individuals may cause omissions and to evaluate the traditional reliance on household respondents as the source of information about who lives in a household. Evidence from the Living Situation Survey is used to identify categories of persons most at risk of omission due to disagreements and to suggest further directions for research to improve the rostering process in household surveys.

Introduction

Who lives in a sampled household is the first and most fundamental thing a survey or census must determine if interviews with its members are to represent a population completely. Yet, little systematic attention has been given to sources of measurement error in compiling household rosters. In this article I examine, and challenge, the assumption made in surveys that household residents are reliably and consistently reported by household respondents.

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In the decennial census and its household surveys, the Census Bureau attempts to enumerate each person at his or her "usual residence," defined as the place where a person lives and sleeps most of the time. This basic principle is modified by several exceptions; for example, commuter workers who work at one place during the week and travel home on the weekends are to be enumerated where they live during the week, not at home.

Although residence is not in question for most people, listing errors are not negligible. They account for about one-third of gross census omissions and contribute to relatively high net undercounts of males, young adults, and minorities (Hogan 1993). An estimated 8.47 percent of black males were missed in the 1990 decennial census, compared with 2.97 percent of black females, 1.94 percent of nonblack males, and .61 percent of nonblack females, according to demographic comparisons of census results with birth and death records and other independent data (Robinson et al. 1993). Undercoverage rates are higher in household surveys than they are in the census (Hainer et al. 1988). Differential undercounts distort census and survey estimates of the characteristics of U.S. households and persons, because the people who are missed are different in many ways from those who are counted.

Research suggests that causes of within-household coverage errors are rooted both in motivational factors and in the complexity and fluidity of households and individuals' living situations. A pioneering ethnographic study sponsored by the Census Bureau in a low income block in Brooklyn provided the first direct evidence that household members were deliberately concealed from the Census Bureau (Valentine and Valentine 1971). Rosters obtained by the ethnographers were compared with rosters obtained independently by Census Bureau interviewers at the same addresses. The study concluded that female respondents in economically marginal households deliberately concealed men from the government when disclosure of a male's presence could jeopardize household income from public assistance or from his participation in the underground economy or illegal activities. They found "a widely shared perception of officialdom and powerful external agencies in general as exploitive and oppressive in their relations with the community" and noted that "the minimum of trust required for people to reveal themselves just does not exist in this context" (Valentine and Valentine 1971, p. 19). A qualitative study by Hainer (1987) in a low-income black neighborhood affirmed the importance his informants placed on consistently presenting an "official" version of household membership that matches records of government bureaucracies and that commonly omits men in order to protect household resources. Based on ethnographic work in Philadelphia, Anderson (1990) suggested fear and a sense of powerlessness as motives for concealment,

noting his informants' intimidation by "paper" and fear of being "written up," especially by anyone in an official capacity.

Motivational barriers to enumeration are compounded by the complex and fluid structure of many households that do not necessarily fit the Census Bureau's concept of a "usual residence." Anthropologists (Stack 1974; Hainer 1987) working in black inner-city communities described large, loosely structured domestic units, including both kin and fictive kin with rather flexible and fluid living arrangements, spread over and moving among several addresses. People may "live" at two or more addresses simultaneously and enter and leave households frequently. For many people it is unclear which of several addresses is their "usual residence," so they may be enumerated more than once or not at all. Others may slip between the cracks because they have no "usual residence" at the time of the census. (The census form instructs respondents to include on the roster anyone staying there on census day who had no usual residence, but it is doubtful that such people are reliably included.) An ethnographic evaluation of behavioral causes of undercount in the 1990 Census confirmed that undercounts were higher in complex and irregular households that contained unrelated or mobile individuals, ambiguous household members, or two or more nuclear families, or that were formed for the sole purpose of sharing rent or living expenses (de la Puente 1993). Much of the research has focused on undercounts in poor households, but affluent households as well may be uncertain where to count part-time residents, such as college students, commuter workers, or children in joint custody arrangements.

Gerber (1990, 1994) applied methods of cognitive anthropology to investigate how respondents decide where someone lives. She found that when life circumstances are complex and ambiguous, her respondents used various criteria to determine residency, including a person's intentions and agreements, where a person keeps belongings or receives mail, financial contributions, permanence of attachment, and other criteria that may conflict with official residency rules (Gerber 1990). Gerber suggests that "an individual's social role or life-style may lead respondents to conclude, in the absence of complete social information, that certain persons 'must have' residences of their own somewhere" (1994, p. 3). Their calculations and inferences may lead respondents to exclude "marginal" people who should be included, so that persons without a stable residence may not be included in census rosters in any of the places they stay. Hainer (1987) finds that an individual's status as a household member is conditional on provision of resources and acceptable role performance. Household membership thus may be a subject of disagreement, with some "members" not acknowledging others. As Hainer et al. note, "This implies there is good reason to suspect the reliability of proxy responses

from these households. If individuals do not agree among themselves as to who is a 'member' of their family, proxies cannot be fully reliable about household rosters" (1988, p. 528).

The qualitative research is consistent with research by Fay (1989) and by Ellis (1994, 1995), showing that a marginal relationship within a household predicts inconsistent reporting of a person in the census and in independent interviews in the same household. People who are marginally related to the household (e.g., boarders) were much more likely to be left off either the census or the survey roster than people who are more central. Ethnographic research (e.g., Rodriguez and Hagan 1991) suggests that unrelated individuals may be left off census rosters because they are not regarded as "family" or as part of the core household.

Hypotheses

This article explores the possibility that a person's lifestyle may give rise to erroneous assumptions and disagreements that result in omissions. Specifically, I examine the following hypotheses:

HYPOTHESIS 1. Respondents infer usual residence (in part) from an individual's presence in a household.

This hypothesis may seem obvious, since usual residence is defined as where a person lives and sleeps most of the time. However, much qualitative research has suggested that presence is not the overriding criterion, and may not be considered at all, in determining residence.

HYPOTHESIS 2. Absence leads to the (sometimes incorrect) inference that an individual has another residence and, hence, reduces the reliability of reporting and increases the level of disagreements.

Hypothesis 2 is based on Gerber's (1994) finding that lack of knowledge and transiency led her respondents to erroneously assume that a person "must have" a residence of his own somewhere else. Sweet and Alberti's (1994) analysis also suggests that disagreement is indicative of ambiguous or atypical residence situations, such as more than one residence, staying at a place most of the week while working, and so on.

HYPOTHESIS 3. Related individuals are more likely to be reported as usual residents and are reported more consistently with fewer disagreements and hence better coverage.

Hypothesis 3 is consistent with prior research showing that unrelated persons are more likely to be omitted from rosters, perhaps because they are not regarded as core household members.

HYPOTHESIS 4. Disagreements introduce bias because they differentially increase the risk of omission of persons (erroneously) reported as nonresidents.

The Living Situation Survey

My analysis is based on an experimental pilot survey that was designed to improve coverage by building more inclusive household rosters. The Living Situation Survey (LSS) was fielded by Research Triangle Institute in summer of 1993. Personal interviews were conducted in 999 households oversampled in areas with high concentrations of minorities and renters. The response rate was 79.5 percent. Sample weights were designed to produce nationally representative results (Lynch et al. 1993).

1. Step 1 in the survey was to ask household respondents to identify all persons with any attachment to the sample households during a 2-month reference period. Extensive cues targeted tenuously attached persons and others at risk of omission. Respondents were asked to mention persons who stayed the previous night, for whom a room is kept, who received mail or messages there or had a key, and so on. The probes were developed based on evidence about undercounts, as well as cognitive and anthropological research on conceptual and belief systems related to residency (Gerber 1990, 1994). A total of 3,537 people were listed on rosters—almost one person added per household on average, compared to the census.
2. Step 2 was to determine the residence of each person on the list. Household respondents were asked if this was the person's usual residence, "where he/she lives and sleeps most of the time." The question was worded to provide respondents the official definition of usual residence. About three-quarters of the persons listed on rosters were reported to be usual residents of the sample units, and one-quarter of the persons listed on rosters lived somewhere else.
3. Step 3 was to follow up a subsample of the persons listed on rosters to collect more information about their living situations. Casual visitors, who had stayed a week or less and lived somewhere else, were not followed up. The most stable residents were subsampled. Follow-up interviews were conducted for 85 percent of eligible individuals ($N = 452$)—one-third of the interviews were with the individuals themselves, and the remainder with knowledgeable proxies.¹

1. For children 12 years old and younger, the interviewer identified the adult who was most knowledgeable about where the child stayed during the reference period and interviewed that person. Older residents were interviewed in person if they were home at the time of the household interview, or by telephone if a phone number was available. Phone numbers and addresses were obtained for nonresidents, and they or their proxies were interviewed in person or by telephone. Late in the field period, on a case-by-case basis,

The weighted mean of 3.62 persons listed per housing unit in the LSS is significantly greater than the mean of 2.63 persons per occupied housing unit in the 1990 census for the nation as a whole (Sweet 1994). When nonresidents were screened out, the mean number of usual residents per housing unit in the LSS was still higher than the census for all race/ethnicity categories but was significantly higher only for the total population (2.76 versus 2.63) and for Hispanics (4.12 versus 3.28) (Sweet 1994). Thus, compared to the census, there was a 38 percent increase in the number of people listed on rosters per household but only a 5 percent increase in the number of usual residents per household. More probing was needed to list persons with tenuous attachments than those who were more attached: on average, 1.06 probes were needed to list the most strongly attached individuals, compared to 4.6 probes needed to elicit reports of persons with very weak attachments to sample households (Martin 1996). The additional probing was especially effective at identifying young minority males, who were less likely to be mentioned in response to the initial, more standard probes (Sweet 1994).

Results

In the follow-up interviews (step 3), individuals (or their proxies) were asked to identify which, if any, of the places they had stayed during the reference period were their usual residences; they were permitted to identify more than one usual residence, though few did. These data allow us to compare individuals' reports with reports given by the original household respondents.

I analyze interrelations among five variables:

1. AGREEMENT between household respondent and follow-up reports of an individual's usual residence (1 = reports agree; 2 = disagree)

Reports agree if both indicate an individual was a usual resident of the sample unit, or both reports indicate he was not a usual resident. They disagree if one but not the other reports him as a resident.

2. HOUSEHOLD RESPONDENT'S REPORT of an individual's usual residence (1 = is a UR; 2 = is not a usual resident of the sample unit)
3. PRESENCE in the household (1 = present in the household all but 0–7 nights of the 2-month reference period; 2 = away more than 1 week)

interviewers were allowed to use the household respondent as a proxy for sample individuals ages 13 and older if they had spent all but 0–7 nights in the sample unit.

4. PROXY STATUS of follow-up interview (1 = follow-up interview conducted with individual him/herself; 2 = knowledgeable proxy)
5. RELATIONSHIP to household respondent (1 = related; 2 = unrelated)

Table 1 presents the cross-classification of the first four variables. In each 2×2 subtable, the row variable is the household respondent's (HHR) report of a person's usual residence (UR), and the column variable is agreement between reports. Percentages in each subtable sum to 100, with standard errors in parentheses. Data are weighted to national household totals and to account for oversampling, nonresponse, and subsampling of persons for follow-up.

Table 1, part A shows that the usual residence of persons who were continuously present was not in question. Their status was clear-cut and known to them and to others reporting for them. Household respondents declared all such individuals to be usual residents, and reports given in follow-up interviews almost universally agreed for both self and proxy interviews.² This result corresponds to the assumption that rosters are reliably reported and unproblematic and, in fact, describes most of the population (about 159 million people, according to this survey).

The situation was quite different for individuals who spent more than 1 week away from the household, shown in part B of table 1. These individuals were less likely to be reported as usual residents, and disagreements were more common. The level of disagreement was also higher for proxy interviews: 6 percent of self-reports and 17 percent of proxy reports contradicted the original report. Persons for whom proxy interviews were given may have lifestyles that make them both difficult to locate and to report about accurately.

The level of agreement is higher for persons initially claimed as usual residents. To illustrate, under part B, the odds on the second, self-report agreeing were over 30 to 1 ($= 78.7/2.6$) for persons initially reported as usual residents but only 4.8 to 1 ($= 15.4/3.2$) for those reported as nonresidents. That is, a report that an individual was a usual resident was 6.3 times more likely to be confirmed in a second interview than a report that a person was not a resident. A more extreme version of the same pattern also holds for proxies. The evidence suggests that survey researchers should not place too much confidence in respondents' reports that someone does not live in a household.

To test the alternative hypotheses and select the best-fitting model that describes these data, table 2 compares various models fitted to the cross-

2. It should be kept in mind that the two interviews were not necessarily independent. In particular, some proxy interviews were given by the household respondent—the data do not identify which ones—that would tend to inflate agreement for proxy reports for individuals absent no longer than a week during the reference period.

Table 1. Agreement between Initial (Household Respondent) and Follow-up Reports of Individual's Residence, by Individual's Presence in Household and Proxy Status of the Follow-up Interview

Presence of Individual in Sample Unit	Self Report in Follow-up Interview			Proxy Report in Follow-up Interview		
	Agrees with Initial Report (%)	Disagrees with Initial Report (%)	Total	Agrees with Initial Report (%)	Disagrees with Initial Report (%)	Odds on Agreement
A. Absent no more than a week (HHR's report in original interview):						
Individual is a usual resident	99.9 (.1)	.1 (.1)	100.0	99.9 (.1)	.1 (.1)	1000
Individual is not a usual resident	0	0	0	0	0	...
Total	99.9	.1	100.0 (N = 43)	99.9	.1	100.0 (N = 96)
B. Absent more than a week (HHR's report in initial interview):						
Individual is a usual resident	78.7 (5.5)	2.6 (1.4)	81.4 (5.5)	66.3 (10.2)	1.8 (.9)	36.8
Individual is not a usual resident	15.4 (5.6)	3.2 (1.7)	18.6 (5.5)	16.6 (6.6)	15.4 (11.6)	1.1
Total	94.2 (2.7)	5.8 (2.7)	100.0 (N = 115)	82.9 (12.0)	17.1 (12.0)	100.0 (N = 188)

NOTE.—Excluded are household respondents and casual visitors who were not followed up in individual interviews. Standard errors are given in parentheses.

Table 2. Comparisons of Log-Linear Models Fitted to the Cross-Classification of AGREEMENT, HHR REPORT, PRESENCE, PROXY, RELATIONSHIP

A. Goodness of Fit				
Model	Marginals Fitted	χ^2	df	p
1	All 2-ways	.27	16	.35
2	All 2-ways, except PRESENCE $\times$ AGREEMENT	.66	17	.22
3	All 2-ways, except PRESENCE $\times$ HHR REPORT	4.46	17	.00
4	All 2-ways, except AGREEMENT $\times$ HHR REPORT	3.49	17	.00
5	All 2-ways, except AGREEMENT $\times$ RELATIONSHIP, HHR REPORT $\times$ RELATIONSHIP			
6	PRESENCE $\times$ AGREEMENT, PRESENCE $\times$ HHR REPORT, AGREEMENT $\times$ HHR REPORT, AGREEMENT $\times$ PROXY, PRESENCE $\times$ RELATIONSHIP, PROXY $\times$ RELATIONSHIP	.16	18	.39
		-.02	20	.46
B. Comparisons of Models				
Models Compared	Parameters Tested	Difference		
		L ²	df	p
1 versus 2	PRESENCE $\times$ AGREEMENT	2.62	1	.004
1 versus 3	PRESENCE $\times$ HHR REPORT	5.28	1	<.0001
1 versus 4	AGREEMENT $\times$ HHR REPORT	4.05	1	<.0001
1 versus 5	AGREEMENT, HHR REPORT $\times$ RELATIONSHIP	-1.41	2	>.5
1 versus 6	AGREEMENT $\times$ RELATIONSHIP, HHR REPORT $\times$ RELATIONSHIP, HHR REPORT $\times$ PROXY, PRESENCE $\times$ PROXY	-.10	4	.44

NOTE.—Jackknifed Pearson χ^2 used to evaluate model fit. Jackknifed likelihood ratio $\chi^2(L^2)$ used to compare models (see Fay and Turner 1989).

classification of all five variables: the four variables shown in table 1, as well as RELATIONSHIP. Log linear models were fitted using RASCHPLX (Fay and Turner 1989), which takes into account the complex sample design. In part A, the goodness of fit of various models is assessed by comparing observed frequencies with frequencies expected under the model. In part B, models were compared by calculating the improvement or deterioration in fit in the likelihood ratio chi-square test statistic (L^2) (Goodman 1971).

Model 1, which includes all pairwise associations among the five variables, provides an acceptable fit to the data. Models 2–4 each drop one of the associations—PRESENCE $\times$ AGREEMENT, PRESENCE $\times$ household respondent's report (HHR REPORT), and AGREEMENT $\times$ HHR REPORT—to separately test their contributions to the fit of model 1. The first three comparisons in part B show that all three of these associations are statistically significant and cannot be dropped from model 1. As indicated by the comparison of models 1 and 2, AGREEMENT is significantly related to PRESENCE. As shown in table 1 (and in table 3, below), the association is positive, supporting the hypothesis that residence is more consistently reported for individuals who are continuously present. This result suggests that as absences increase in frequency or duration, household respondents' assessments of residency become more difficult and are more likely to be contradicted in a second interview. Comparison of models 1 and 3 indicates a strong association between PRESENCE and HHR REPORT, supporting the hypothesis that presence is used by household respondents as a criterion for determining residency. Thus, individuals who were absent for over 1 week were less likely to be reported as usual residents (see table 1). In addition, there is a direct positive association between HHR REPORT and AGREEMENT, as indicated by the comparison of models 1 and 4. That is, a household respondent's report that an individual is a usual resident is much more likely to be confirmed in a second, follow-up interview than a report that an individual is not a usual resident. Finally, comparison of models 1 and 5 tests the influence of RELATIONSHIP on HHR REPORT and on AGREEMENT. Although these variables have been found in previous research to correlate significantly with omission and inconsistent reporting, the effects of relationship are not significant when other variables in the model are controlled. Thus, these two associations may be dropped from the final model without a significant loss of fit.

Model 6 is the best-fitting model describing these data, in the sense that adding additional effect parameters does not significantly improve its fit, nor can any of the effect parameters be dropped without a significant loss of fit. Parameter estimates for model 6 are presented in table 3. The λ parameters are analogous to coefficients of linear regression, where the object is to predict the distribution of frequencies among the 32 cells

Table 3. Parameter Estimates for a Log Linear Model Fitting Five Variables

Effect	Parameter Estimates		
	λ	SE	Standardized Value
Grand mean	4.057	.330	12.284
AGREEMENT between HHR and follow-up reports (1 = reports agree; 2 = disagree)	2.010	.377	5.330
HHR REPORT of individual's usual residence (1 = UR; 2 = not UR)	1.795	.202	8.892
Individual's PRESENCE (1 = continuously present; 2 = away more than 1 week)	-2.148	.340	-6.317
PROXY STATUS (1 = self for follow-up interview; 2 = proxy)	-.421	.198	-2.12
RELATIONSHIP to HHR (1 = related; 2 = nonrelated)	1.647	.184	8.944
AGREEMENT $\times$ HHR REPORT	.819	.214	3.825
HHR REPORT $\times$ PRESENCE	1.888	.146	12.900
AGREEMENT $\times$ PRESENCE	.836	.318	2.631
AGREEMENT $\times$ PROXY	.458	.198	2.311
PRESENCE $\times$ RELATIONSHIP	.354	.172	2.06
PROXY $\times$ RELATIONSHIP	-.135	.170	-.792
Jackknifed Pearson $\chi^2 = -.02, df = 20, p = .46$			

formed by the cross-classification of the five variables. The single parameter shown for each effect is the difference between the effect for the first level of a variable and the average effect. (This parametric form is found in Goodman 1978 and elsewhere.) Statistically significant effects have values that are at least twice their standard errors, as indicated by standardized values greater than two.

Model 6 includes positive parameters corresponding to the three pairwise associations discussed above, between AGREEMENT and PRESENCE (standardized $\lambda = 2.631$), PRESENCE and HHR REPORT (standardized $\lambda = 12.9$), and HHR REPORT and AGREEMENT (standardized $\lambda = 3.825$). It includes an association between PRESENCE and RELATIONSHIP (standardized $\lambda = 2.057$), indicating that related persons tend to be present more than unrelated ones, but does not include direct associa-

tions between RELATIONSHIP and HHR REPORT, or RELATIONSHIP and AGREEMENT. Thus, the effects of relationship are indirect: because unrelated individuals are more likely to be absent, they are less likely to be regarded as usual residents or to be reported consistently. The best-fitting model also includes a significant association between PROXY status and AGREEMENT, indicating that self-respondents are more likely to agree with household respondents' reports than proxies are. (As shown in table 1, part B, reports agree for 94 percent and 83 percent of self and proxy follow-up interviews, respectively.) A (nonsignificant) parameter reflecting an association between PROXY and RELATIONSHIP must also be included in the model to control for the joint distribution of these independent variables. Model 6 fits the data very well, with jackknifed Pearson $\chi^2 = -.02$ on 20 *df*, and $p = .46$.

These findings may help explain the dynamics of undercoverage within households. I assert that persons reported as nonresidents in the original interview and as residents in the second interview (those in the "disagree" column and the "not a usual resident" row of each subtable of table 1) are at risk of omission. This inference seems clear-cut when the second interview is self-reported, since we assume an individual is most knowledgeable about his own living situation.³ When both interviews are given by proxy, it is less clear which is closer to truth. Regardless, disagreements between a household respondent and a second proxy are likely to result in omissions. Both respondents are reporting, in effect, "he lives there, not here" and neither is counting the person as a household resident. Thus, the survey yields an estimate of about 4.6 million persons at risk of omission nationally. They are balanced by an estimated 882,000 persons (corresponding to the individuals in the "disagree" column and "usual resident" row of the subtables in table 1) who are at risk of erroneous inclusion. The net difference implies a potential net undercoverage of 3.7 million people. Although based on a small sample and large standard errors, this result suggests that errors made in compiling household rosters do not balance out but may lead to the exclusion of large numbers of individuals whose residence is in question.

The reasons why respondents did not consider individuals listed on rosters to be household members provide additional insight into the types of situations giving rise to disagreements and potential omissions. Table 4 shows that the most common reason given was that the person lives or stays somewhere else, and/or was visiting or staying temporarily in the household (67 percent), followed by reasons related to financial, legal, or family relationships (22 percent). Slightly over one-tenth of individuals were reported to have moved. The reason given predicts how likely the

3. Individual self-reports as well as household respondent reports may be inconsistent with official residence rules, however; see Sweet and Alberti (1994) for an examination of the consistency of reporting with official rules.

Table 4. Reasons Given by Household Respondents for Not Considering Rostered Individuals to be Household Members

“What are the reasons you do not consider NAME to be a member of this household?”	Percent of Total	Percent of HHRs’ Reports Confirmed in Follow-up Interviews
Reasons related to financial, legal, or household arrangements or family relationships (e.g., “he doesn’t pay rent,” “pays own bills, buys own food, has separate area,” “we are not married,” “I filed for divorce today. She went off to see her brother in California and left me behind. I don’t need her”)	22 (16.15)	98
Person lives or stays somewhere else; visits or stays temporarily (e.g., “has a place of his own,” “she has somewhere else to stay,” “in jail,” “on a military base,” “he was visiting on vacation,” “he was only staying to finish school,” “‘cause he comes and goes as he pleases,” “he doesn’t sleep here on a regular basis,” “just in or out,” “only receives mail here”)	67 (23.66)	73
Person has moved or will move (e.g., “has married and moved away,” “got kicked out,” “will be moving”)	11 (6.96)	0
Total individuals reported as nonmembers by HHRs	100	30
Unweighted <i>N</i>	77	

follow-up interview is to confirm the household respondent’s report of an individual as a nonmember. The third column of table 4 shows the percentage of individuals in each category who were confirmed in follow-up interviews not to be members of the original sample households. (The association between reason and agreement between the two reports is marginally significant: jackknife Pearson $\chi^2 = 1.24$, $df = 2$, $p < .073$.) The reasons are ordered in table 4 from those most likely to be associated with consistent reporting to reasons associated with greatest inconsistency.

When a household respondent reported an individual not to be a household member for legal or financial reasons or because of family relationships, the follow-up interview almost unanimously confirmed the report. Evidently such arrangements are consensual among potential respondents, including the individuals themselves. A household respondent's report that an individual lived somewhere else and/or was only staying temporarily was confirmed in 73 percent of cases. This result is consistent with the hypothesis that many (here, 27 percent of 67 percent, or 18 percent) household respondent reports that persons reside elsewhere are questionable and may be erroneous. Finally, household respondents' reports that individuals were not household members because they had moved were least likely of all to be confirmed: all such reports were contradicted in follow-up interviews, which claimed that the "movers" were still members of the original households. It seems probable that this extreme result is a fluke of the small sample on which this analysis is based. Nevertheless, this result points to an area where further research is needed; movements in and out of households may be far less reliably reported than survey researchers have tended to assume. Most inconsistent assessments of household membership appear to result from incorrect assumptions that individuals have other residences or from uncertainty or disagreement about whether individuals have moved out or not. (Of the potential omissions, 38 percent were persons reported to have moved, and most of the remainder were said by household respondents to live elsewhere.) Inconsistently reported movers may be people who the household respondent has decided no longer belong in a household, and who are in the process of being thrown or eased out. (The person who "got kicked out" appears to be a case of this sort.) The examples in table 4 also suggest that some determinations of residency were based on plans or expectations of a move in advance of the event, which may also give rise to inconsistent reporting. Survey researchers need to examine further the role of moves in residence determinations, and the causes for the high rates of inconsistency found here.

Conclusions

Surveys almost uniformly accept as reliable respondent reports of who lives in a household. For most American households, this assumption is appropriate. But for persons with uncertain household attachments, household respondents may not be able to provide accurate information and may be biased toward excluding individuals who are not continuously present and about whose living arrangements they have incomplete information. According to results analyzed here, almost 4 million people nationally may be at risk of omission for this reason.

The results are consistent with ethnographic and qualitative research into the household dynamics that result in census and survey omissions. Results of this survey support the results of earlier qualitative research that household respondents may not be knowledgeable about the residential arrangements of all persons staying in their households and may draw incorrect inferences based on observations of a person's lifestyle. The results also suggest an explanation for the well-documented correlation between relationship and coverage error. According to these data, unrelated persons are more vulnerable to coverage error not because of the effects of relationship *per se*, but because unrelated persons tend not to be continuously present in households.

The results point to several areas for future research. First, some situations may strongly indicate likely misreporting. In this survey, persons away for more than 1 week during a 2-month period appear to be at considerable risk of inconsistent reporting and potential omission. The data strongly suggest that a household respondent's report that a person does not live in a sample unit should not be accepted at face value in these circumstances. Knowing this, survey designers may add questions to identify persons whose circumstances are vulnerable to misreporting and follow-up on them separately to confirm their residence.

Second, the data indicate that individual moves out of households warrant more research attention; they appear not to be the definite, consensually defined events survey researchers have assumed. Mobility is a major cause of attrition in longitudinal surveys and is associated with high rates of omission in the census. The finding that moves were very inconsistently reported in this survey should be replicated with a larger sample, and the possible sources of misreporting explored more fully. Do persons misremember when moves occurred or perceive their timing differently, giving rise to inconsistent reports? Are some household respondents reporting on the basis of wishful thinking that a troublesome individual "moved out"? Possibly, young adults in the process of moving out and establishing a separate residence also maintain (at least in their own minds) residence in the parental household as well. This possibility is consistent with Schwede and Ellis's (1994) finding that inconsistent assessments of household membership were significantly higher for young adults than for other age groups.

Likewise, the relationship between an individual's presence in a household and reports of his residence there needs further investigation. Other analysis of these data suggests that the pattern of mobility, rather than the amount of time spent away from the household, influences household respondents' residency determinations. Bates and Gerber (1998) identify a pattern of repeat visits to the same place, indicating strong attachments to a second household, and a "floating" pattern of mobility among multi-

ple households as patterns that create ambiguity and make it difficult for household respondents to determine usual residence.

Third, the results support the hypothesis that uncertain or ambiguous residential arrangements may be an important cause of omissions. Results of this analysis are not fully consistent with the conclusion reached by Tourangeau and his colleagues (1997) that concealment, not confusion, is the major explanation for roster omissions. The LSS reveals there are many people whose living situations give rise to inconsistent reports by different respondents. Yet these people were not concealed, though they were not reported as residents.

Fourth, the results point to the need to evaluate current rostering methods and adapt them to changes in Americans' living situations and family arrangements. Perhaps, as Gerber (1994) and Tourangeau et al. (1997) also suggest, new counting and residency rules are needed to link individuals to households. Government and private surveys alike have tended to treat the household roster almost as an administrative detail prior to the survey proper and have not treated the roster as a measurement problem. In some government surveys, terminology and procedures have remained basically the same for 50 years; it is time research was devoted to this neglected topic to update and improve this fundamental component of all surveys: the household roster.

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